

Decoding the Shopper: Customer Behavior & Retention Analytics

An End-to-End Retail Analytics Project

Python • PostgreSQL (SQL) • Power BI

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1. Project Overview

Retail businesses generate vast volumes of transactional data every day, yet the real strategic value lies not in the data itself but in the patterns hidden within it. This project undertakes a complete, end-to-end analysis of 3,900 customer transactions to understand how demographics, product preferences, discounts, and service choices come together to shape customer spending and loyalty.

The workflow mirrors a professional data analytics pipeline: raw data is cleaned and enriched in Python, structured business questions are answered through SQL queries in PostgreSQL, and the resulting insights are brought to life in a fully interactive, custom-designed Power BI dashboard spanning two pages - an executive Overview and a drill-through Deep Dive.

2. Problem Statement

A retail organization sought to better understand the forces driving customer purchasing behavior in order to strengthen sales performance, improve customer satisfaction, and build long-term loyalty. Leadership had observed shifting purchase patterns across different customer segments, product categories, and fulfillment channels, and needed clarity on which factors - promotional discounts, product reviews, seasonality, or payment preferences - most meaningfully influenced buying decisions and repeat business.

The guiding business question addressed throughout this project is:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize its marketing and product strategies?”

3. Dataset

The analysis is built on a transactional dataset comprising 3,900 individual customer purchases spread across 18 attributes, capturing three broad dimensions of the customer journey:

- Customer demographics - age, gender, geographic location, and subscription status
- Purchase details - item purchased, product category, purchase amount, season, size, and color
- Shopping behavior - discount usage, promo code usage, prior purchase history, purchase frequency, review rating, and shipping method

A minor data quality gap was identified during initial profiling: 37 missing entries in the review rating field, later addressed during the cleaning phase described in Section 5.

4. Tools and Technologies

- Python (Pandas, Matplotlib, Seaborn) - data cleaning, feature engineering, and exploratory analysis
- PostgreSQL - relational database for structured, query-based business analysis
- pgAdmin - database administration and CSV data import
- Power BI Desktop - interactive dashboard design and DAX-based measures
- Jupyter Notebook - analysis and documentation environment

- Git & GitHub - version control and portfolio publication

5. Data Cleaning and Preparation

Before any analysis could take place, the raw dataset was systematically profiled and cleaned in Python to ensure it was reliable and analysis-ready:

1. Loaded the raw dataset into a Pandas DataFrame and profiled its structure using `df.info()` and `df.describe()` to understand data types, ranges, and distribution.
2. Identified and resolved 37 missing values in the review rating field by imputing the median rating within each product category, preserving category-level rating patterns rather than applying a single global average.
3. Standardized every column name to snake_case formatting for consistency and downstream compatibility with SQL.
4. Engineered two new features: an `age_group` column created by binning raw ages into meaningful demographic brackets, and a `purchase_frequency_days` column derived from the customer's stated purchase cadence.
5. Audited the `discount_applied` and `promo_code_used` fields for redundancy - confirmed they captured overlapping information - and removed the duplicate `promo_code_used` column to keep the dataset lean.
6. Extended the standard cleaning workflow with an outlier audit on `purchase_amount` using the interquartile range (IQR) method, visualized with a boxplot, to confirm the spend distribution contained no extreme, analysis-distorting values.
7. Ran a correlation analysis across the key numeric fields (`age`, `purchase_amount`, `review_rating`, `previous_purchases`) using a heatmap, to check for underlying relationships between customer attributes and spending before moving into SQL.
8. Exported the fully cleaned dataset to CSV and loaded it into a PostgreSQL database through pgAdmin, establishing the single source of truth used for every SQL query that followed.

6. Exploratory Data Analysis - Key Insights

- Purchase amounts are tightly distributed between \$20 and \$100, averaging \$59.76, with the IQR outlier check confirming no abnormal high- or low-spend transactions distorting the dataset.
- Review ratings show no strong linear correlation with purchase amount - customers do not necessarily spend more simply because they rate items more favorably, suggesting satisfaction and spend are driven by different underlying factors.
- Only 27% of the customer base holds an active subscription, yet subscriber spend closely tracks non-subscriber spend - pointing to a conversion opportunity rather than a value gap.

7. Data Analysis Using SQL

With the cleaned dataset loaded into PostgreSQL, fifteen structured business questions were formulated and answered - the original ten core questions plus five additional questions introduced to

explore dimensions of the dataset (location, season, payment method, and purchase frequency) that were not covered in the base analysis.

7.1 Revenue by Gender

Compared total revenue contribution between male and female customers to understand which demographic segment drives the larger share of sales.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

7.2 High-Spending Discount Users

Identified customers who applied a discount yet still spent above the overall average purchase amount - a segment of price-sensitive but high-value shoppers.

	customer_id integer	purchase_amount numeric
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76
19	43	100
20	44	69
21	55	94
22	57	73
23	58	64

7.3 Top 5 Products by Average Rating

Ranked products by their average customer review rating to surface the items customers are most satisfied with.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

7.4 Standard vs. Express Shipping

Compared average purchase amounts between Standard and Express shipping customers to assess whether shipping preference correlates with spend.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

7.5 Subscribers vs. Non-Subscribers

Compared customer counts, average spend, and total revenue across subscription status to quantify the value of the subscriber base.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

7.6 Discount-Dependent Products

Identified the five products with the highest share of discounted purchases, flagging categories most reliant on promotions to drive sales.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7.7 Customer Segmentation (New / Returning / Loyal)

Classified every customer into New, Returning, or Loyal segments based on prior purchase count, producing a clear view of the loyalty funnel.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

7.8 Top 3 Products per Category

Ranked the top three best-selling products within each product category using a window function, revealing category-level bestsellers.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

7.9 Repeat Buyers and Subscription Likelihood

Examined whether customers with more than five prior purchases were more likely to hold an active subscription.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

7.10 Revenue by Age Group

Quantified each age group's contribution to total revenue, highlighting which demographic bands matter most commercially.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

7.11 Top Revenue-Generating Locations

Ranked the top ten customer locations by total revenue - a geographic dimension not explored in the original analysis - to identify where regional marketing investment would be most effective.

	location text	total_revenue numeric	total_customers bigint
1	Montana	5784.00	96
2	Illinois	5617.00	92
3	California	5605.00	95
4	Idaho	5587.00	93
5	Nevada	5514.00	87
6	Alabama	5261.00	89
7	New York	5257.00	87
8	North Dak...	5220.00	83
9	West Virgi...	5174.00	81
10	Nebraska	5172.00	87

7.12 Revenue and Spend by Season

Compared total revenue and average spend across seasons to inform a seasonal promotions and inventory calendar.

	season text	total_orders bigint	avg_purchase numeric	total_revenue numeric
1	Fall	975	61.56	60018.00
2	Spring	999	58.74	58679.00
3	Winter	971	60.36	58607.00
4	Summer	955	58.41	55777.00

7.13 Revenue by Payment Method

Identified which payment method is used most often and which generates the highest total revenue, relevant to checkout and processing-fee strategy.

	payment_method text	total_customers bigint	total_revenue numeric	avg_purchase numeric
1	Credit Card	671	40310.00	60.07
2	PayPal	677	40109.00	59.25
3	Cash	670	40002.00	59.70
4	Debit Card	636	38742.00	60.92
5	Venmo	634	37374.00	58.95
6	Bank Transfer	612	36544.00	59.71

7.14 Spend by Purchase Frequency

Compared average spend and total revenue across purchase-frequency bands (e.g., weekly, monthly, annually) to identify high-value but infrequent customers worth re-engaging.

	frequency_of_purchases text	total_customers bigint	avg_purchase numeric	total_revenue numeric
1	Every 3 Months	584	60.08	35088.00
2	Annually	572	60.17	34419.00
3	Quarterly	563	59.98	33771.00
4	Bi-Weekly	547	60.69	33200.00
5	Monthly	553	59.33	32810.00
6	Fortnightly	542	59.05	32007.00
7	Weekly	539	58.97	31786.00

7.15 Review Rating vs. Repeat Purchases

Tested whether customers who leave higher review ratings also tend to have more prior purchases, directly addressing the business question of whether reviews drive loyalty.

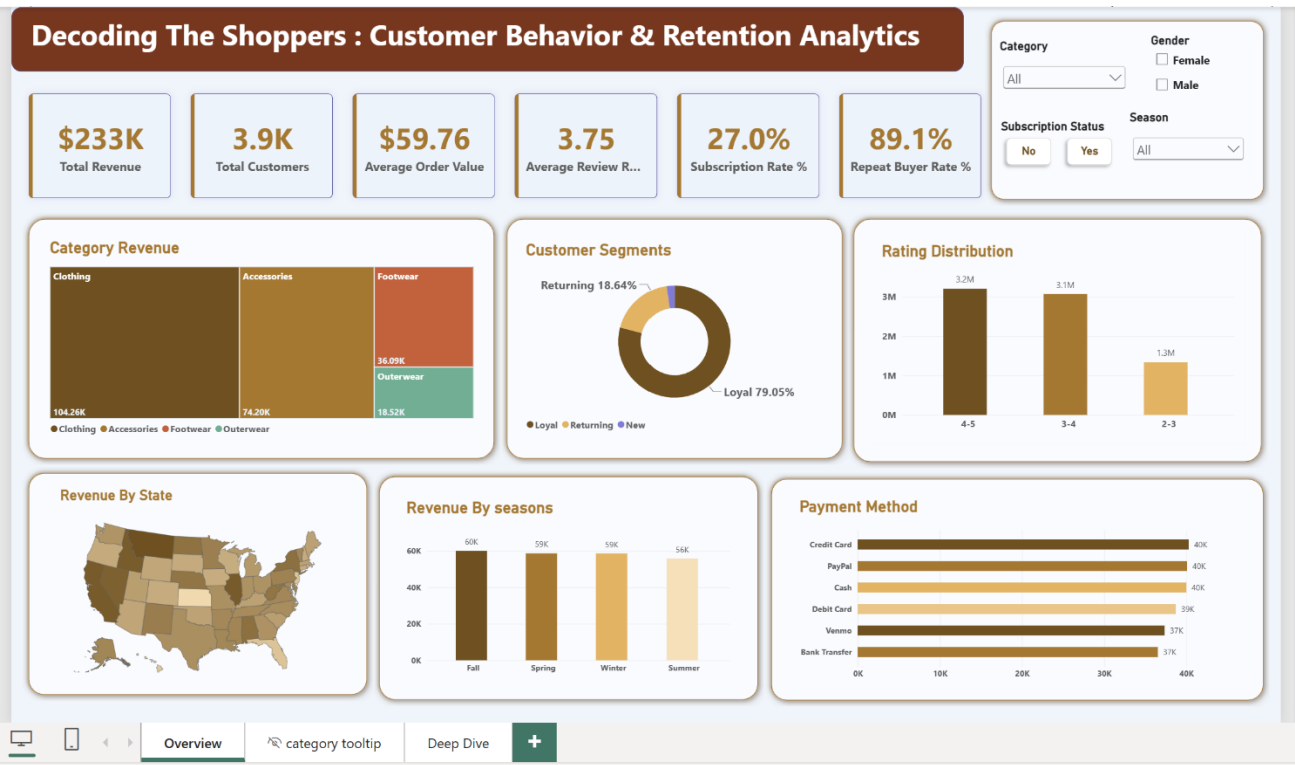
	rating_bucket text	avg_previous_purchases numeric	total_customers bigint
1	Medium (3-4)	25.47	1767
2	High (above 4)	25.35	1453
3	Low (below 3)	25.05	680

8. Dashboard in Power BI

The insights generated through Python and SQL were consolidated into a custom-designed, two-page interactive Power BI dashboard - built to move beyond a static report and let stakeholders explore the data themselves.

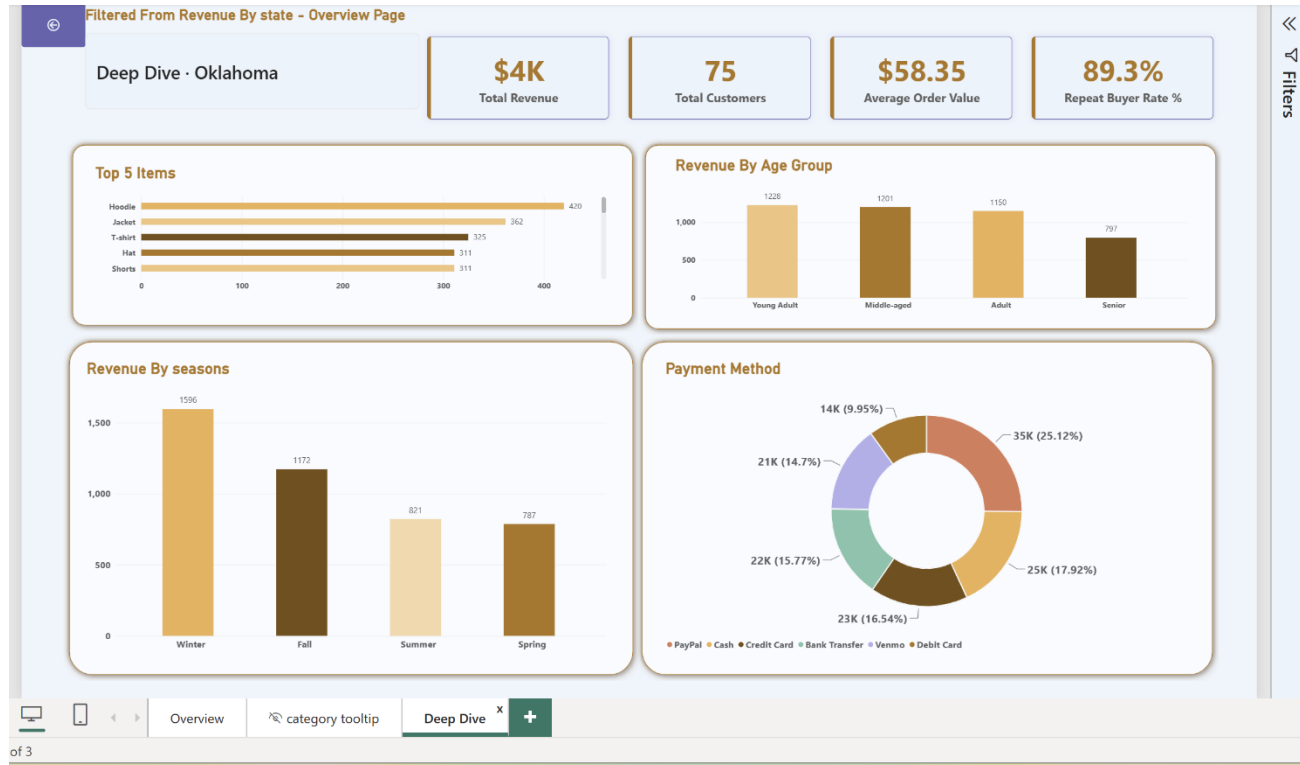
8.1 Overview Page

The Overview page presents the full customer base at a glance: six key-performance-indicator cards (Total Revenue, Total Customers, Average Order Value, Average Review Rating, Subscription Rate, and Repeat Buyer Rate), a category revenue treemap, a customer-segment donut chart, a review-rating distribution chart, a filled map showing revenue by state, and supporting breakdowns by season and payment method. Slicers for gender, category, season, and subscription status allow real-time filtering across every visual on the page.



8.2 Deep Dive Page

The Deep Dive page is a drill-through destination: clicking any state on the Overview page's map instantly navigates here, automatically filtered to that state, and reveals its top five products, payment-method split, age-group breakdown, and seasonal pattern - with a dynamic breadcrumb title confirming the selected context and a back button to return to the Overview page.



9. Results and Conclusion

9.1 Key Findings

- Male customers generated roughly twice the revenue of female customers, indicating a significant demographic imbalance in current sales.
- Only 27% of customers are subscribed, despite subscriber spend closely matching non-subscriber spend - a clear, low-risk upsell opportunity.
- The customer base is heavily loyalty-weighted: 3,116 customers fall into the Loyal segment, compared to just 701 Returning and 83 New - retention is strong, but new-customer acquisition appears underinvested.
- Categories such as Hats and Sneakers show discount rates approaching 50%, raising a margin concern that warrants a pricing and promotions review.
- Express shipping customers spend marginally more on average than Standard shipping customers, suggesting a mild willingness to pay for convenience.
- Review ratings show limited correlation with repeat-purchase behavior, suggesting loyalty is driven more by factors like pricing, convenience, or brand habit than by satisfaction scores alone.

9.2 Business Recommendations

- Launch a targeted subscription campaign to convert high-spending non-subscribers, given their spend already matches the subscriber base.
- Introduce structured loyalty incentives aimed specifically at moving Returning customers into the Loyal segment.
- Reassess discount policy on high-discount-dependency products (e.g., Hats, Sneakers) to protect margins without suppressing volume.
- Feature top-rated and top-selling products more prominently in marketing and merchandising campaigns.
- Concentrate marketing spend on the highest-revenue locations and seasons identified through the SQL analysis, rather than applying a uniform national or year-round strategy.
- Invest in top-of-funnel acquisition channels, given the unusually thin New customer segment relative to the Loyal base.
- Use review ratings as an early customer-service signal - proactively reaching out to customers who leave low ratings before they churn.
- Negotiate or promote the highest-revenue payment method at checkout to reduce transaction friction and processing costs.

10. Author and Contact

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GitHub Repository: <https://github.com/Vipparti-Vinay/customer-behavior-retention-analysis-python-sql-powerbi>

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